

Assignment-2 (DS)

1. Implement **Stack ADT** and **Queue ADT** using template classes in C++. Demonstrate the operations insertion, deletion, and searching with suitable examples. Analyze the **time complexity** and **space complexity** of each operation.
2. Explain the representation of a **linear list** using arrays and linked lists. Compare it with **skip list representation** in terms of structure, searching efficiency, insertion, and deletion operations. Also discuss how **recursion** can be applied in data structure operations with an example.